DO NOW — RECALL

Dijkstra: shortest paths I

Name: _____ Class: _____ Date: _____

1. Why does quicksort exclude the pivot from its recursive calls?

2. Define a weighted edge.

3. What is the first index of an array?

4. A–B costs 3 and B–C costs 5. What is cost of A→B→C?

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

The pivot is already in its final sorted position.

2. Model answer

A connection between vertices with an associated cost or distance.

3. Model answer

0.

4. Model answer

8.

Vocabulary — know these terms

tentative distance — Best known cost to an unsettled vertex.

settled — Vertex whose shortest distance has been finalised.

relaxation — Improve a neighbour's distance via the current vertex.

predecessor — Previous vertex on the best-known route.

non-negative weight — Edge cost of zero or greater.

NEW LEARNING

Read, annotate and explain

Initialise from the start

Distance(A)=0, all others infinity. Relax A's edges to set B=4 and C=2.

Always choose smallest tentative distance

C=2 settles before B=4. Through C, B improves to 3.

Keep updating neighbours

Settling B changes D from 10 to 8; settling D changes E from 12 to 10.

Reconstruct from predecessors

E←D←B←C←A. Reverse these arrows for A→C→B→D→E, total distance 10.

Dijkstra's requirement

Edge weights must be non-negative. A visited/settled node's shortest distance is then final.

Explain one step in your own words, using the correct technical terms.

I DO — WORKED EXAMPLE

Follow the worked method

Find the shortest P→S route using non-negative weights.

Predict the next state

P gives tentative Q=3 and R=7.

Reveal and explain

Settle Q=3, improving R to 5 and discovering S=9.

Reveal and explain

Settle R=5, improving S to 6.

Reveal and explain

Predecessors S←R←Q←P give P→Q→R→S, cost 6.

WE DO — GUIDED PRACTICE

Try → Discuss → Improve

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 1

A–B=4 and A–C=2. Starting Dijkstra at A, state the tentative distances of A, B and C after relaxing A's neighbours.

Question 2

On edges S–A=3, S–B=7, A–B=2, B–T=2 and A–T=8, find the shortest S→T path with full distance reasoning.

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 3

Explain why Dijkstra must not be relied upon when negative-weight edges are possible, and contrast tentative and settled distances.

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

Dijkstra always chooses the next vertex by its smallest edge weight.

How confident are you? Circle one: Confident / Getting there / Need help

Explain your choice, identifying one concept you can now use independently.

GREEN PEN — Correct and explain

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

Next step: revisit the model or a relevant revision-guide section if needed.